University Hackathon 2026 Submission

FlowTransit Project

Theme: PEOPLE
Category: Mobility

1. Executive Summary

Master-planned townships are designed for peaceful, green living, but often fail at their perimeters due to severe highway exit bottlenecks during peak hours. **FlowTransit** is an AI-powered Demand-Responsive Transit (DRT) solution designed to eliminate these localized traffic jams. By aggregating passenger demand and granting our shared electric shuttles **Transit Signal Priority (TSP)** at critical intersections, we transition communities from single-occupancy vehicle gridlock to efficient, zero-emission mass micro-transit.

2. The Problem Statement

Imagine moving into a highly marketed, sustainable township. You find the internal roads calm, but your daily commute is ruined by a brutal reality: the single, one-kilometer drive out to the main highway turns into a 45-minute standstill every morning.

The root cause is structural inefficiency—thousands of residents are driving single-occupancy vehicles toward the exact same bottleneck simultaneously. Traditional navigation apps (like Waze or Google Maps) fail here; they attempt to route drivers through narrow residential lanes (shortcuts), which destroys the safety and quiet of the neighborhood, while still ultimately merging into the same saturated highway exit.

The Core Insight

We cannot solve a physical bottleneck by merely redirecting cars into smaller streets. We must **remove cars from the bottleneck entirely** by grouping passengers into high-efficiency vehicles that never get stuck in the line.

15:1	-20%	0
Car Replacement Ratio	Queue Length Reduction	Emissions (100% EV)

3. Target Audience

Our solution serves three distinct demographics within the township ecosystem:

- **The Daily Commuters:** Professionals who waste up to an hour daily just trying to exit the neighborhood to reach their offices or main highway networks.
- **Students & Non-Drivers:** Residents who require safe, reliable first/last-mile connections to nearby public transit hubs (e.g., MRT/LRT) without relying on parents for drop-offs.
- **The Broader Community:** All residents who currently suffer from noise, pollution, and the severe safety hazards caused by external drivers using their local residential lanes as shortcuts.

4. System Architecture



5. The FlowTransit Solution

FlowTransit shifts the paradigm from "routing cars around traffic" to "prioritizing high-capacity transit." Our solution integrates three technological pillars:

A. Demand-Responsive Routing (DRT)
Instead of running empty buses on fixed, inefficient routes, our AI predicts peak demand and dispatches EV shuttles only where they are needed. Residents book rides via a mobile app, and the algorithm instantly aggregates passengers heading to the same highway exit or transit hub.

B. "Virtual Stops" to Protect Neighborhoods
Rather than sending heavy shuttles down narrow residential alleys to pick people up at their doorsteps, the AI calculates safe "Virtual Stops" located on main internal roads within a 3-minute walk of the passengers. This keeps the residential zones perfectly quiet and safe.

C. Transit Signal Priority (TSP) "Green Wave"
This is our ultimate differentiator. As a fully loaded shuttle (carrying 15 passengers) approaches the congested main exit, IoT sensors communicate with the smart traffic light system. The system grants a "Green Wave" (extending the green light), allowing the shuttle to bypass the massive queue of single-occupancy cars.

6. The User Journey

The experience is designed to be frictionless, taking less than 10 seconds to initiate a commute:

1. **Request:** Resident opens the FlowTransit app and selects "Morning Commute to MRT Station."
2. **Match & Walk:** The AI matches them with a shuttle arriving in 5 minutes and directs them to a Virtual Stop just 150 meters away.
3. **Board & Ride:** The resident boards the electric shuttle along with 8 other neighbors heading the same way.
4. **Bypass:** Approaching the exit bottleneck, the passengers receive a notification: *"TSP Activated: Green Wave Secured."* The shuttle clears the intersection without stopping.

7. Alignment with the 6 Core Dimensions

1 Relevant

Directly solves the exact real-world mobility bottleneck faced by residents in master-planned townships (like City of Elmina), fulfilling the PEOPLE challenge.

2 Impactful

Creates highly measurable outcomes: replacing 15 cars with 1 electric shuttle instantly cuts exit queue lengths, saves commuter hours, and drastically reduces neighborhood CO_2 emissions.

3 Innovative

Unlike standard GPS apps that merely reroute cars into smaller streets, we integrate dynamic dispatching with physical urban infrastructure (Traffic Light overriding).

4 Feasible

Utilizes existing technologies—standard EV passenger vans, proven DRT algorithms, and API-enabled traffic lights—requiring zero new heavy road construction.

5 Scalable

As a software-first solution, the platform can be deployed rapidly across any developer's portfolio of townships facing similar perimeter traffic issues.

6 Evidence-Based

Backed by recent successful Demand-Responsive Transit pilots in Selangor, proving that middle-class residents will adopt shared transit if it offers superior speed and convenience.

8. Implementation Timeline (Pilot Strategy)

To ensure a risk-free rollout, we propose a 3-phase implementation plan tailored for a single test neighborhood before scaling township-wide.

Phase	Duration	Key Objectives & Milestones
Phase 1: Pilot	Months 1 - 3	Deploy 3 EV shuttles in one residential precinct. Map 10 Virtual Stops. Integrate IoT sensors with the main highway exit traffic light.
Phase 2: Validation	Months 4 - 6	Measure traffic reduction at the exit. Gather resident feedback via the app. Fine-tune AI predictive dispatching algorithms based on real-world delays.
Phase 3: Rollout	Months 7 - 12	Scale the fleet to 15 vehicles. Connect multiple precincts. Offer the "Township Mobility API" to neighboring developments to coordinate traffic flows.

9. Conclusion

A connected, fulfilling lifestyle is impossible if residents start their day trapped in gridlock. **FlowTransit** does not just offer another mapping app; it redesigns urban mobility from the ground up. By combining AI predictive routing with intelligent infrastructure, we return time, safety, and fresh air to the community, proving that sustainable, tech-enabled living is both achievable and highly desirable today.

Thank you for reviewing the FlowTransit Proposal.

Designed for the 2026 University Hackathon